Chapter 2

SYSTEM DYNAMICS PROBLEMS WITH **RATE PROPORTIONAL TO AMOUNT**

Metode Penyelesaian Masalah dan Pemodelan

DTETI, FT, UGM



SYSTEM DYNAMICS PROBLEMS WITH RATE PROPORTIONAL TO AMOUNT

Growth and decay (Module 2.2, Module 2.3)

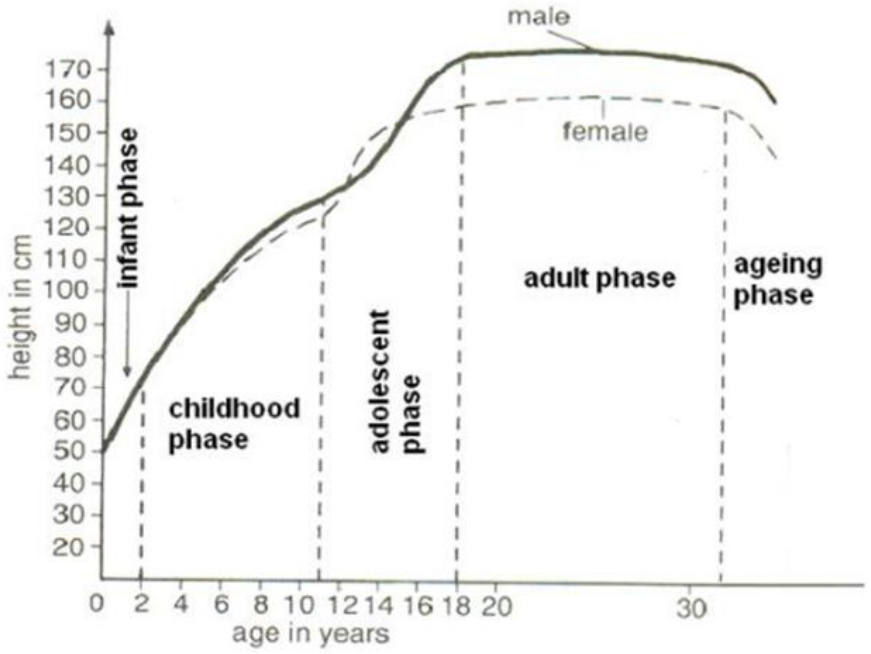
Drug dosage (Module 2.5)

Rate of change is proportional to the amount present.

$$\text{average velocity} = \frac{\Delta s}{\Delta t} = \frac{186.7\ \text{km} - 16.1\ \text{km}}{2\ \text{h} - 0\ \text{h}} = \frac{170.6\ \text{km}}{2\ \text{h}} = 85.3\ \text{km/h}$$

Example

Rate of change: example



Human growth curve

- Growth rate = change of height / change of time
- Growth rate in each phase?



SYSTEM DYNAMICS PROBLEMS WITH **RATE PROPORTIONAL TO AMOUNT**

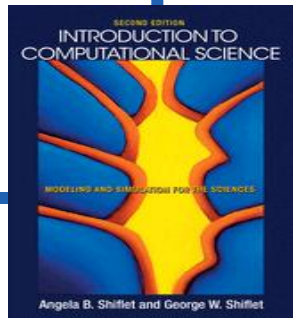
Modeling Growth and Decay

Why modelling growth and decay?

- There are many problems in everyday life that require growth and decay models for prediction, planning and improvement purposes..
  - Population (people, animals, bacteria)
  - Viewers of viral video
  - Radioactivity
  - Toxic waste left in a lake
- There are many other examples of problems, everything that basically concerns the rate of change, whether positive or negative..

- Example:
  - Rate of change of population $\frac{dP}{dt} \propto P$

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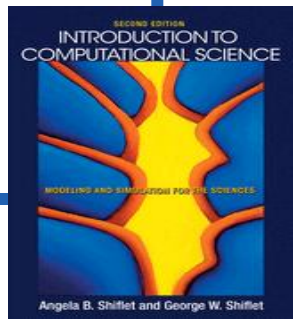
Module 2.2
Unconstrained Growth and Decay

Angela B. Shiflet and George W. Shiflet

Wofford College

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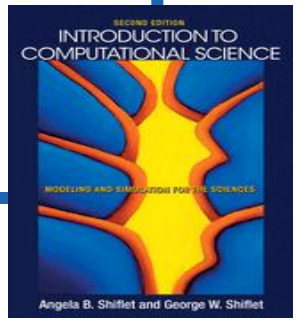
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- Assume we have a population in which
  - no one comes or goes;
  - births and deaths are the only causes of population change.
- There are no constraints such as competition for food or predation.
  - This means there are no restraining factors

    → ***unconstrained*** *growth model*

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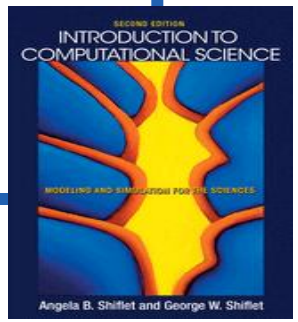
Example of unconstrained growth

- Population growth without constraints
- *Rate of change of population is directly proportional to number of individuals in the population (P)*
- Differential equation $\frac{dP}{dt} = rP$, where ***r*** is growth rate
- $\frac{dP}{dt} = r_1 P - r_2 P = (r_1 - r_2)P$

Malthusian model
(simple exponential growth model)

Model untuk unconstrained growth

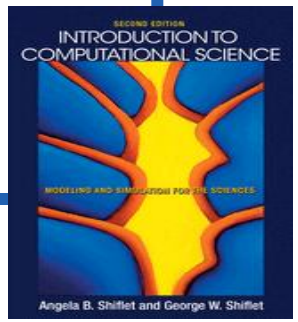
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Example of unconstrained growth

- Differential equation $\frac{dP}{dt} = rP$, where ***r*** is growth rate

Definitions A **differential equation** is an equation that contains one or more derivatives. An **initial condition** is the value of the dependent variable when the independent variable is zero. A **solution** to a differential equation is a function that satisfies the equation and initial condition(s).



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Analytic Solution

- For example, given that growth_rate r=0.1, then

$$\frac{dP}{dt} = 0{,}1P$$

$$\frac{1}{P}dP = 0{,}1dt$$

$$\int \frac{1}{P}dP = \int 0{,}1\,dt$$

$$ln|P| = 0{,}1t + C$$

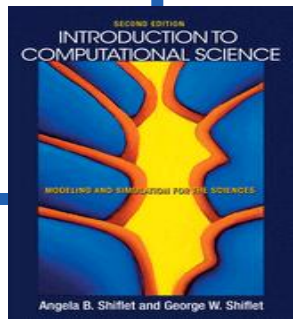
$$e^{\ln|P|} = e^{0.10t+C}$$

$$|P| = e^{0.10t}e^{C} = A\,e^{0.10t}$$

where $A = e^{C}$. Solving for P, we have

$$P = (\pm A)e^{0.10t}$$

$$P = ke^{0.10t}$$



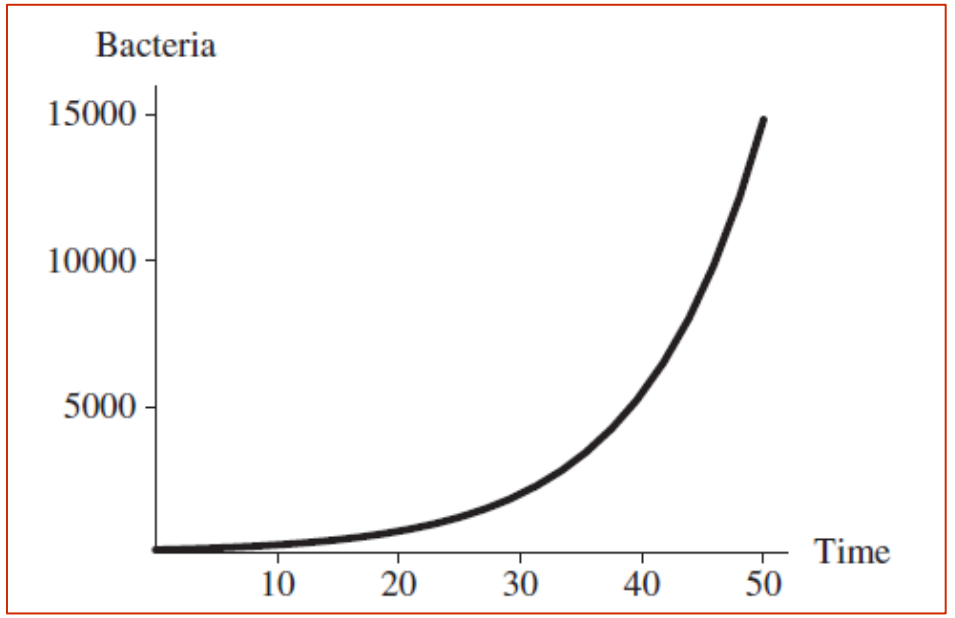
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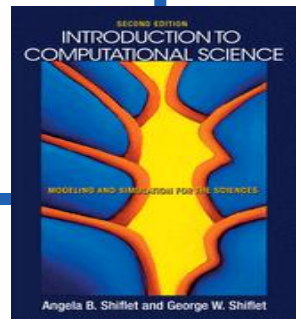
Analytic Solution

- In this example, constan k is the initial population of a particular bacteria. Therefore,

$$P = P_0 e^{rt} = 100\, e^{0.1t}$$

The bacteria population growth is plotted





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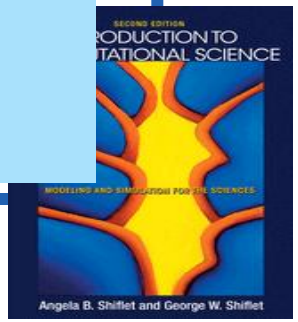
In general, the solution to

$$\frac{dP}{dt} = rP \quad \text{with initial population } P_0$$

is

$$P = P_0 e^{rt}$$

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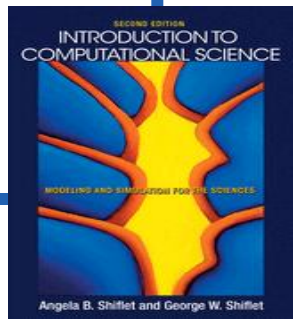
For the same case

- It is known that the initial population of a type of bacteria is 100 and the instantaneous growth rate is 10% = 0.10/hour. So we have a differential equation

  $dP/dt = 0.10Po$ where Po = 100.

  Differential equation

- Can determine with a **computer** algebra system?



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Finite difference equation

Definition A **finite difference equation** is of the following form:

(new value) = (old value) + (change in value)

Such an equation is a discrete approximation to a differential equation.

$$P_t = P_{t-\Delta t} + (P_{t-\Delta t} * r) * \Delta t$$

Finite difference equation

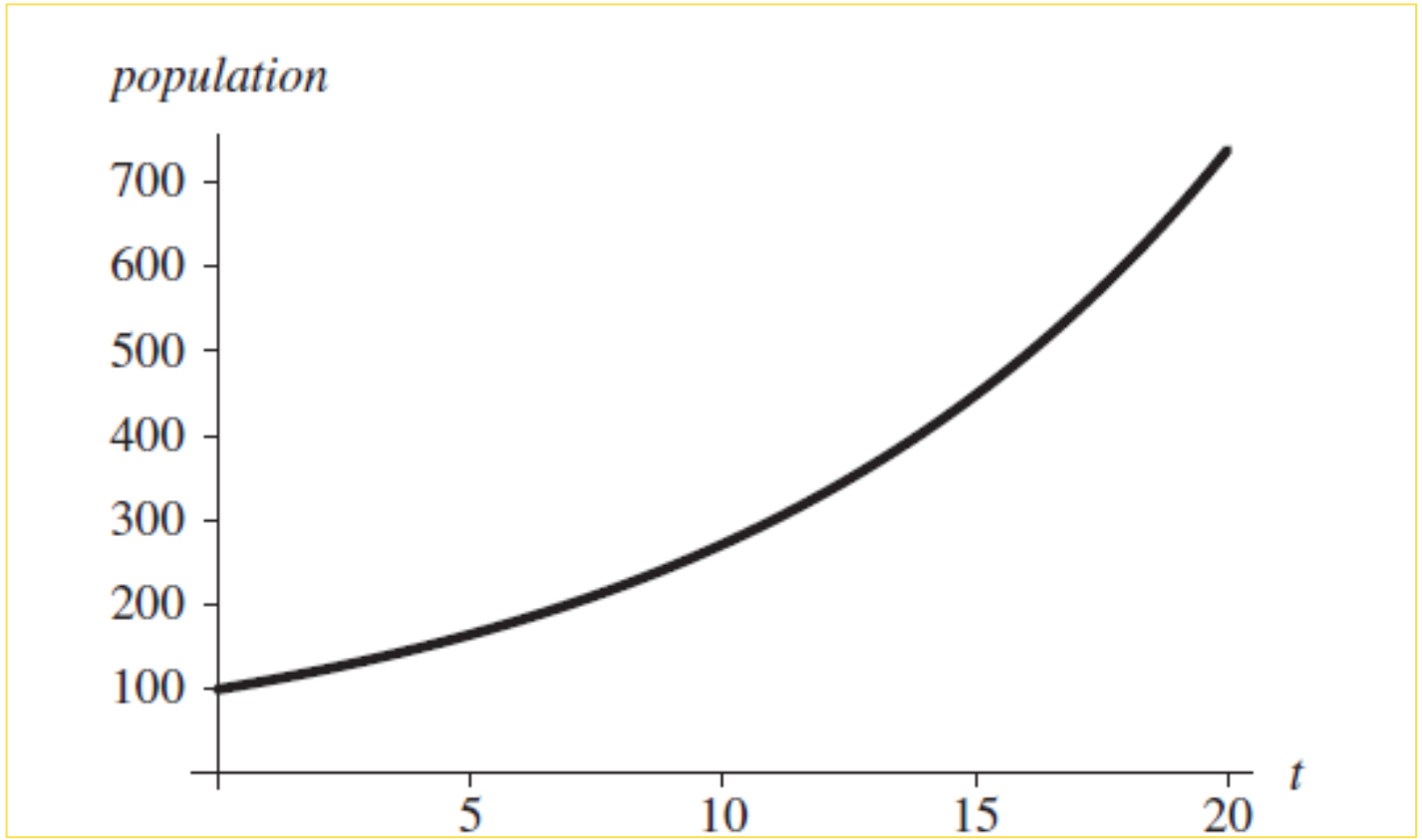
Table 2.2.1
Table of Estimated Populations, Where the Initial Population is 100, the Continuous Growth Rate is 10% per Hour, and the Time Step is 0.005 h

t	$population(t)$	=	$population(t-\Delta t)$	+	$(growth)$	*	Δt
0.000	100.000000						
0.005	100.050000	=	100.000000	+	10.000000	*	0.005
0.010	100.100025	=	100.050000	+	10.005000	*	0.005
0.015	100.150075	=	100.100025	+	10.010003	*	0.005
0.020	100.200150	=	100.150075	+	10.015008	*	0.005
0.025	100.250250	=	100.200150	+	10.020015	*	0.005
0.030	100.300375	=	100.250250	+	10.025025	*	0.005
0.035	100.350525	=	100.300375	+	10.030038	*	0.005
0.040	100.400701	=	100.350525	+	10.035053	*	0.005

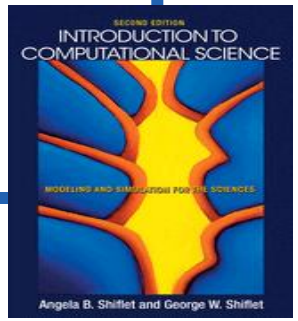
$$P_t = P_{t-\Delta t} + (P_{t-\Delta t} * r) * \Delta t$$

Rule of Thumb Although the simulation takes longer because of more computation, it is usually more accurate to use a small step size (Δt), say, 0.01 or less.

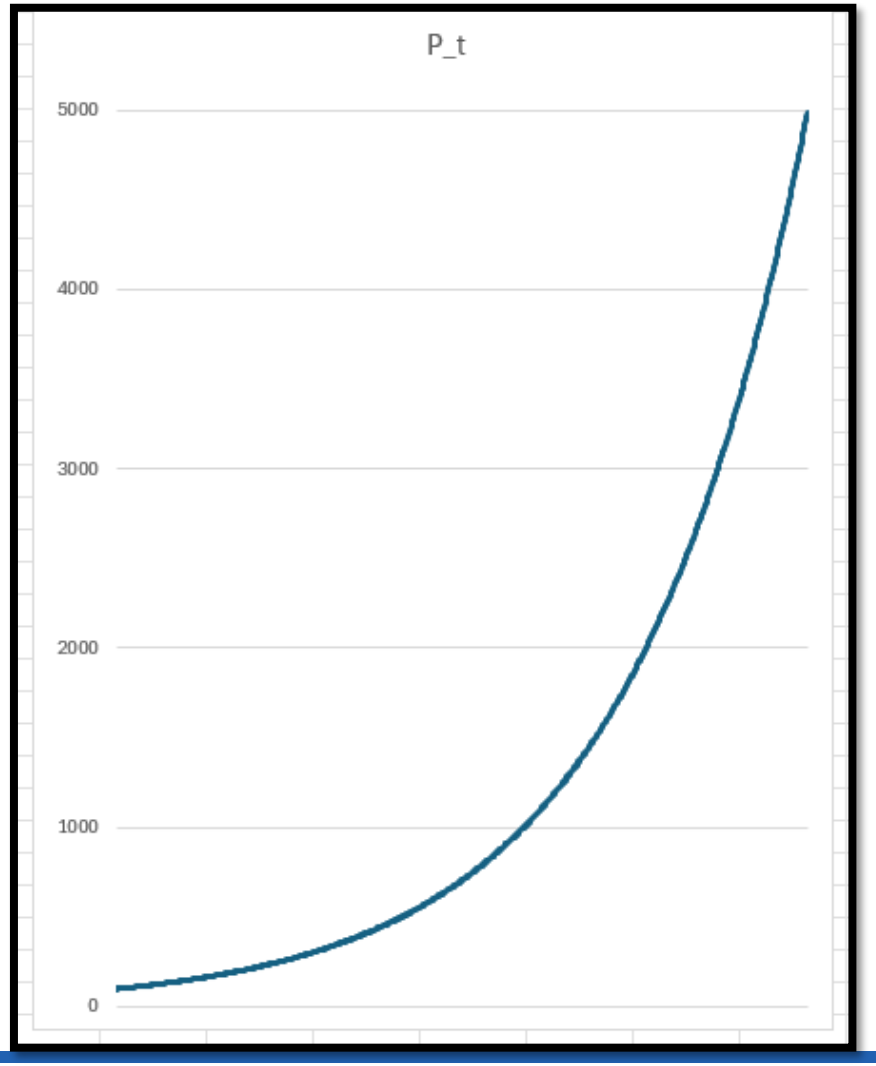
Graph of population versus time (hours)

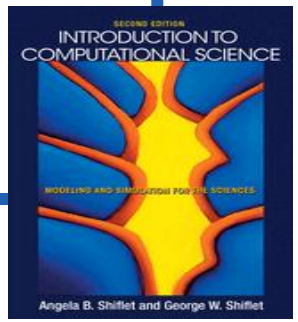


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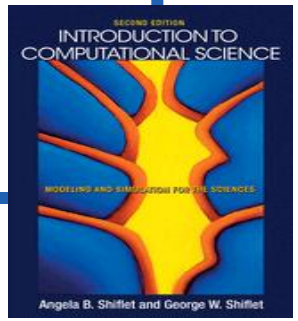


1	Diketahui	r =	0.1		
2					
3	**t**	**P_t**	**P_t-dt**	**G**	**dt**
4	0	100			
5	0.05	100.5	100	10	0.05
6	0.1	101.003	100.5	10.05	0.05
7	0.15	101.508	101.003	10.1003	0.05
8	0.2	102.015	101.508	10.1508	0.05
9	0.25	102.525	102.015	10.2015	0.05
10	0.3	103.038	102.525	10.2525	0.05
11	0.35	103.553	103.038	10.3038	0.05
12	0.4	104.071	103.553	10.3553	0.05
13	0.45	104.591	104.071	10.4071	0.05
14	0.5	105.114	104.591	10.4591	0.05
15	0.55	105.64	105.114	10.5114	0.05
16	0.6	106.168	105.64	10.564	0.05
17	0.65	106.699	106.168	10.6168	0.05
18	0.7	107.232	106.699	10.6699	0.05
19	0.75	107.768	107.232	10.7232	0.05
20	0.8	108.307	107.768	10.7768	0.05
21	0.85	108.849	108.307	10.8307	0.05
22	0.9	109.393	108.849	10.8849	0.05
23	0.95	109.94	109.393	10.9393	0.05





Using modeling and simulation software



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Model representation with several different tools

- Figure a) was created using a modeling tool called Berkeley Madonna ®, b) was created using a modeling tool called STELLA ®, c) was created using Vensim PLE ®, and d) was in the format used in the textbook (Shiflet, 2014).

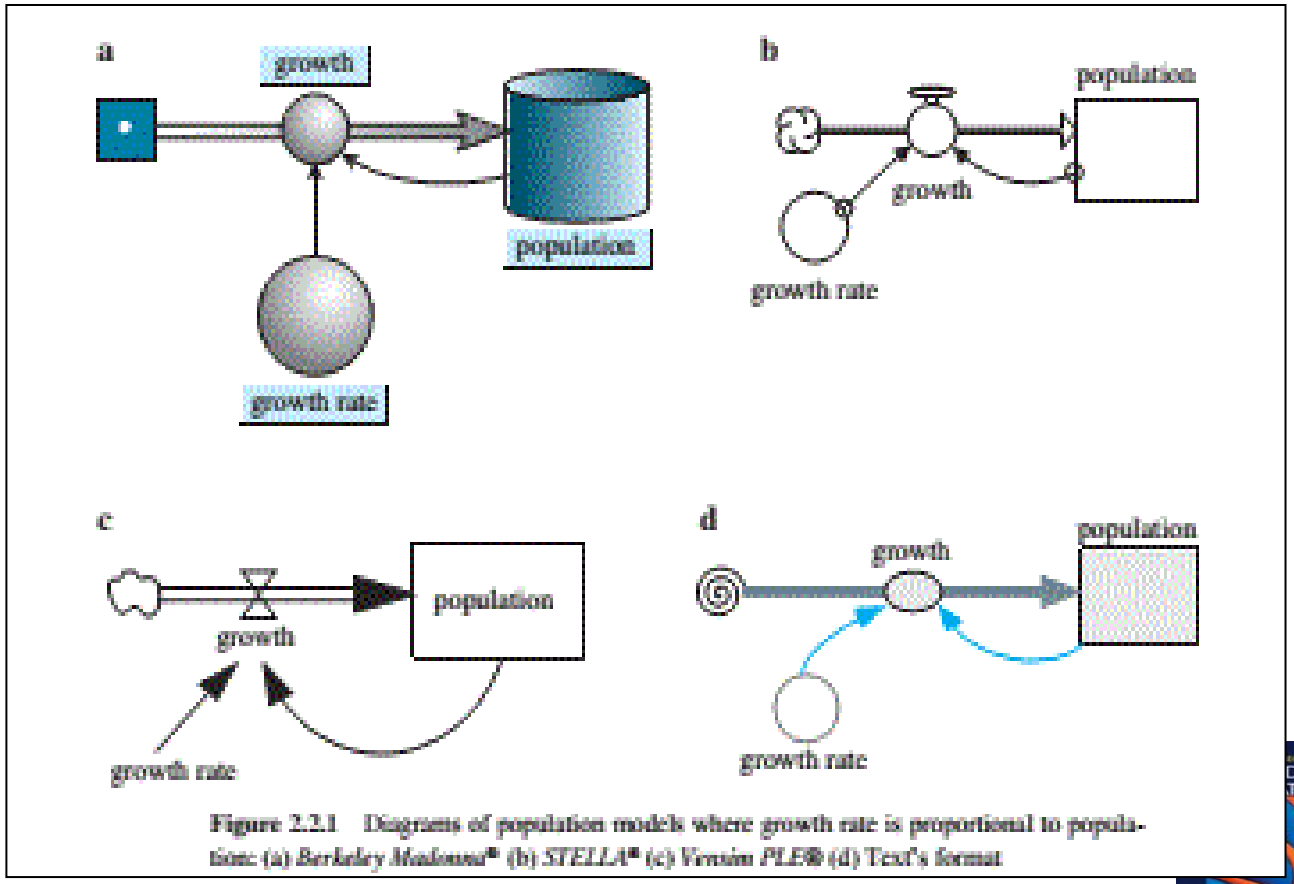
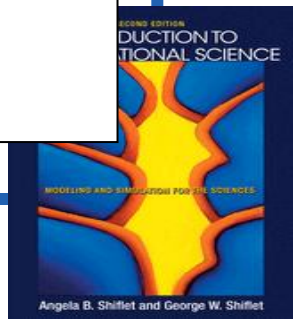


Figure 2.2.1 Diagrams of population models where growth rate is proportional to popula-
tion: (a) *Berkeley Madonna*® (b) *STELLA*® (c) *Vensim PLE*® (d) Text's format



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Examples of modeling and simulation tools: Insight Maker

Home

PRIMITIVE | FLOWS/TRANSITIONS | LINKS | SETTINGS | SAVE | EDIT | SIMULATE

Insight Maker Basics

Click me to Clear this Demo Model

Step 1. Build your Model

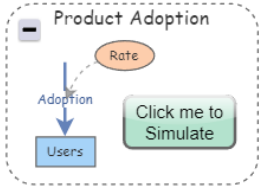
Diagram

Use an extensive set of drawing tools to create causal loop diagrams, flow charts or rich pictures. Build the diagram that communicates your insights and mental models.

This diagram is an Insight Maker model. Feel free to change it. Each picture, textbox or other element is called a "Primitive". You can click and drag primitives to move them or edit their contents in the sidebar.

Simulate

Add equations to your model to simulate future behavior and changes. Use a powerful equation language and large library of built-in functions to simulate System Dynamics or Agent Based Models. View and explore results directly within Insight Maker.



Analyze

Analyze your model results. Study the trends to understand the dynamics of your system. Use Insight Maker's Sensitivity Testing Tool to understand the effect of randomness. Apply Insight Maker's Optimization Tool to maximize or minimize a goal. Utilize Insight Maker's scripting capabilities to take complete control of your analysis.

Step 2. Share your Insights

Insight Maker models are web pages, so sharing them is as easy as emailing a link or posting it to Twitter or Facebook.

If you have your own web page or blog, you can embed Insight Maker models right in your page.

Insight Maker also provides a great way to explain your model: Stories.

Stories create a walkthrough of your model where you can show messages and reveal the model step-by-step.

Welcome to your new Insight!

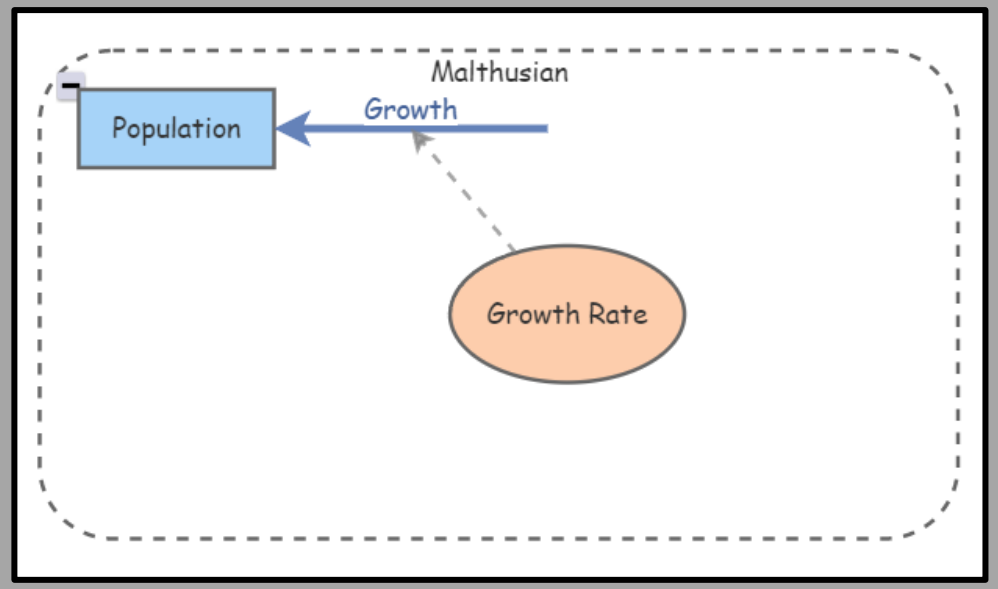
Start diagramming, modelling and simulating.

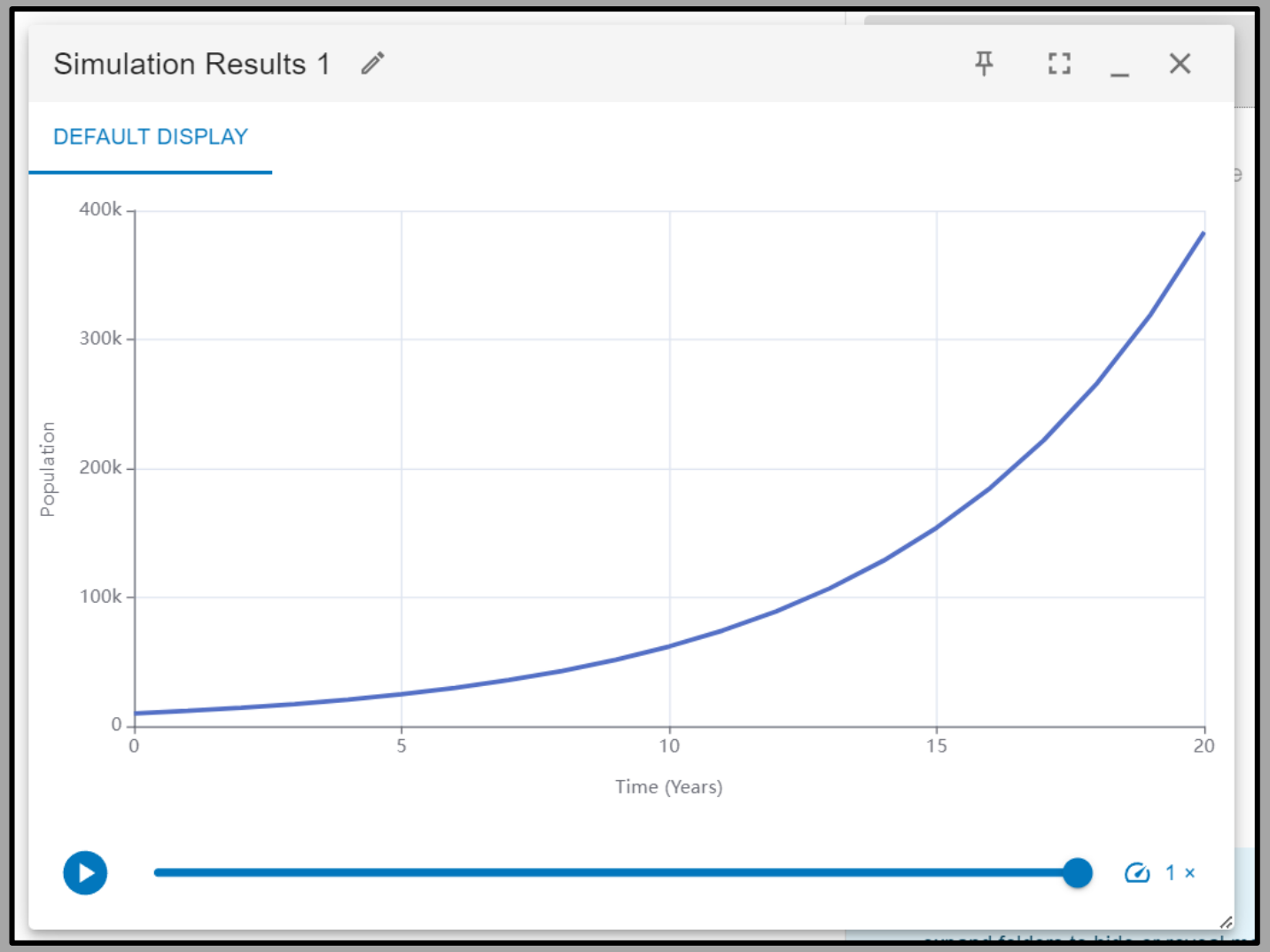
You can experiment with your model now, or take advantage of our help resources.

LEARN MORE | SAVE INSIGHT

Questions or feedback? Email support@insightmaker.com.

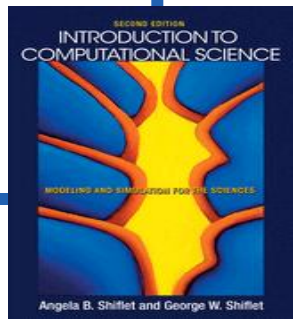
Unconstrained growth model and simulation examples using Insight Maker





System's modeling tool

- Helps to model
- Performs simulation
- What happens at one time step influences what happens at next



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Stock/Box Variable/Reservoir

- Anything that accumulates, buffer, resource
- Examples
  - Population
  - Radioactivity
  - Phosphate
  - Body fat
  - Labor

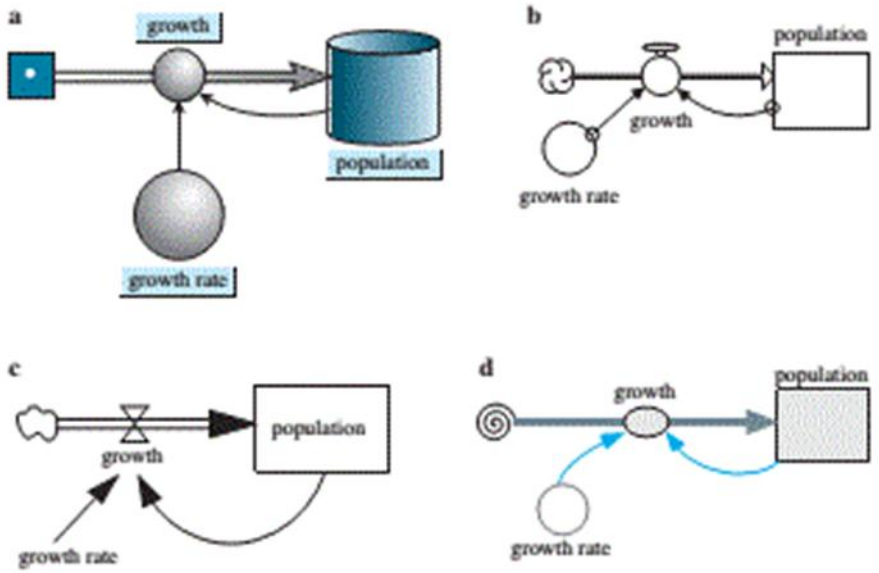
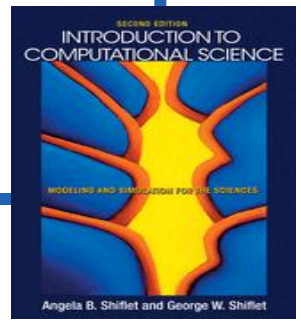


Figure 2.2.1 Diagrams of population models where growth rate is proportional to population: (a) *Berkeley Madonna*® (b) *STELLA*® (c) *Vensim PLE*® (d) Text's format

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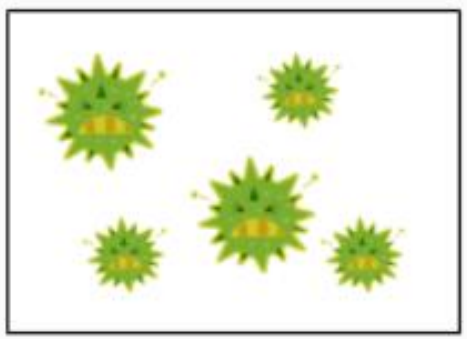
INSIGHT MAKER

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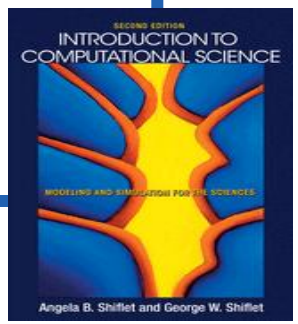
Stocks

A stock is a bucket into which a something can accumulate or be withdrawn from. What the bucket holds is specific to your model; it could be water, or money, or even deer. You can set an initial value or level for the stock that it will take on at the start of the simulation. The initial value may be a simple number or a more complex mathematical expression.





Stocks are flexible. For instance, they can hold people, water, or viruses.



Flow

- Represents activities
- Changes the magnitude of a stock
- Examples
  - Birthing, dying with population
  - Intaking & expending calories with body fat

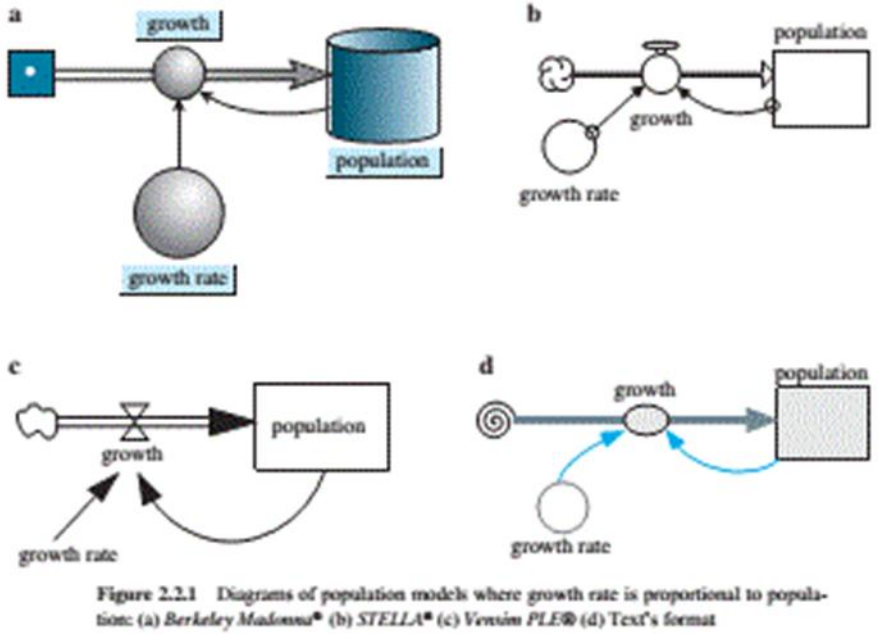
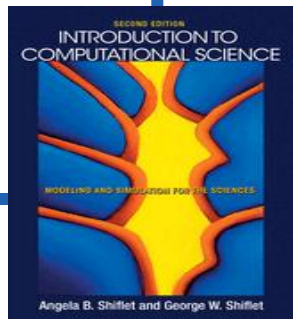


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Converter/Variable/Formula

- Contains equations that generate output for each time period
- Converts inputs into outputs
- Takes in information & transforms for use by another variable
- Examples
  - Growth rate with population & growth
  - Calories in a food

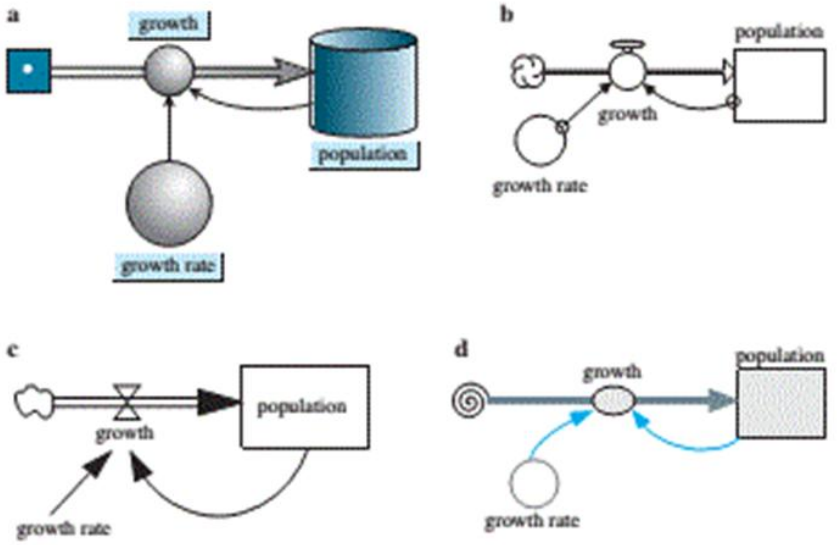


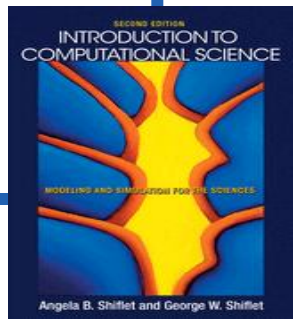
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Connector/Arrow/Arc

- Link
- Transmits information & inputs
- Regulates flows

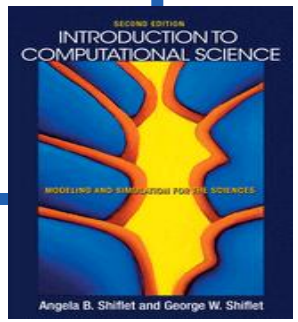
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With system dynamics tool

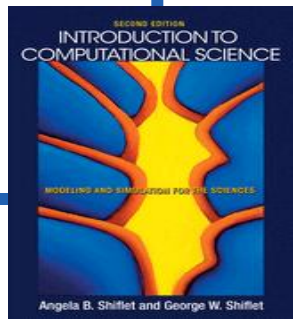
- Enter equations
- Run simulations
- Produce graphs
- Produce tables

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Algorithm for simulation of exponential growth

- initialize *simulationLength*, *population*, *growthRate*. Δt
- *numIterations* ← *simulationLength* / Δt
- for *t* going from 0 to *simulationLength* in steps of size Δt do the following:
  - *growth* ← *growthRate* * *population*
  - *population* ← *population* + *growth* * Δt
  - $t \leftarrow i * \Delta t$
  - display *t*, *growth*, and *population*



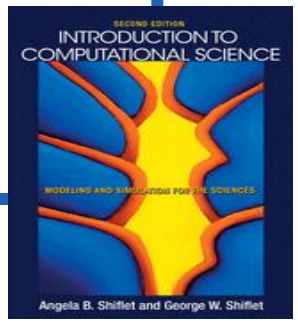
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Algorithm for simulation of exponential growth

Algorithm 1 Algorithm for simulation of unconstrained growth

initialize *simulationLength*
initialize *population*
initialize *growthRate*
initialize length of time step Δt
numIterations $\leftarrow$ *simulationLength*/Δt
for *i* going from 1 through *numIterations* do the following:
    growth $\leftarrow$ *growthRate* * *population*
    population $\leftarrow$ *population* + *growth* * Δt
    $t \leftarrow i * \Delta t$
    display *t*, *growth*, and *population*

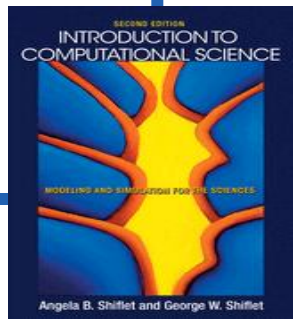
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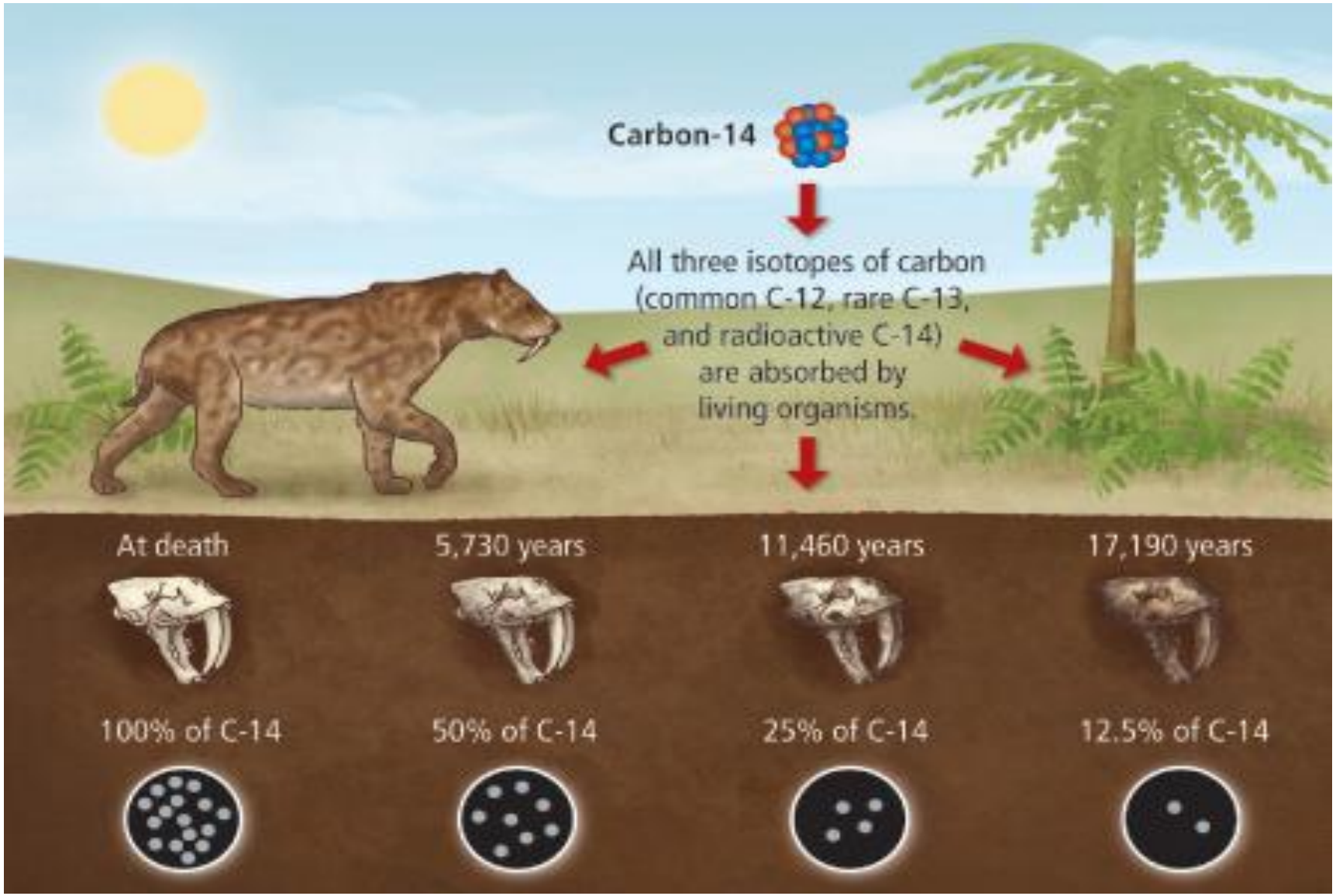


Exponential Decay

- Rate of change of mass of radioactive substance proportional to mass of substance
- Constant of proportionality negative
  - Example: Radioactive carbon-14: -0.000120968
- $dQ/dt = ?$
  - $dQ/dt = -0.000120968Q$
- $Q = Q_0 e^{-0.000120968t}$
- Carbon dating
- The **half-life** is the period of time that it takes for a radioactive substance to decay to half of its original amount Q_0, i.e. 0.5 Q_0.

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Source: https://socratic.org/questions/what-do-scientists-learn-about-fossils-by-using-carbon-14

Example: carbon dating

- What is the half-life of radioactive carbon-14 with a decay constant of -0.000120968 per year?

$$Q = 0{,}5\ Q_0$$

$$0{,}5\ Q_0 = e^{-0{,}000120968t}\ Q_0$$

$$0{,}5 = e^{-0{,}000120968t}$$

$$\ln(0{,}5) = \ln(e^{-0{,}000120968t})$$

$$-0.693 = -0{,}000120968t$$

$$t \approx 5730 \text{ tahun}$$

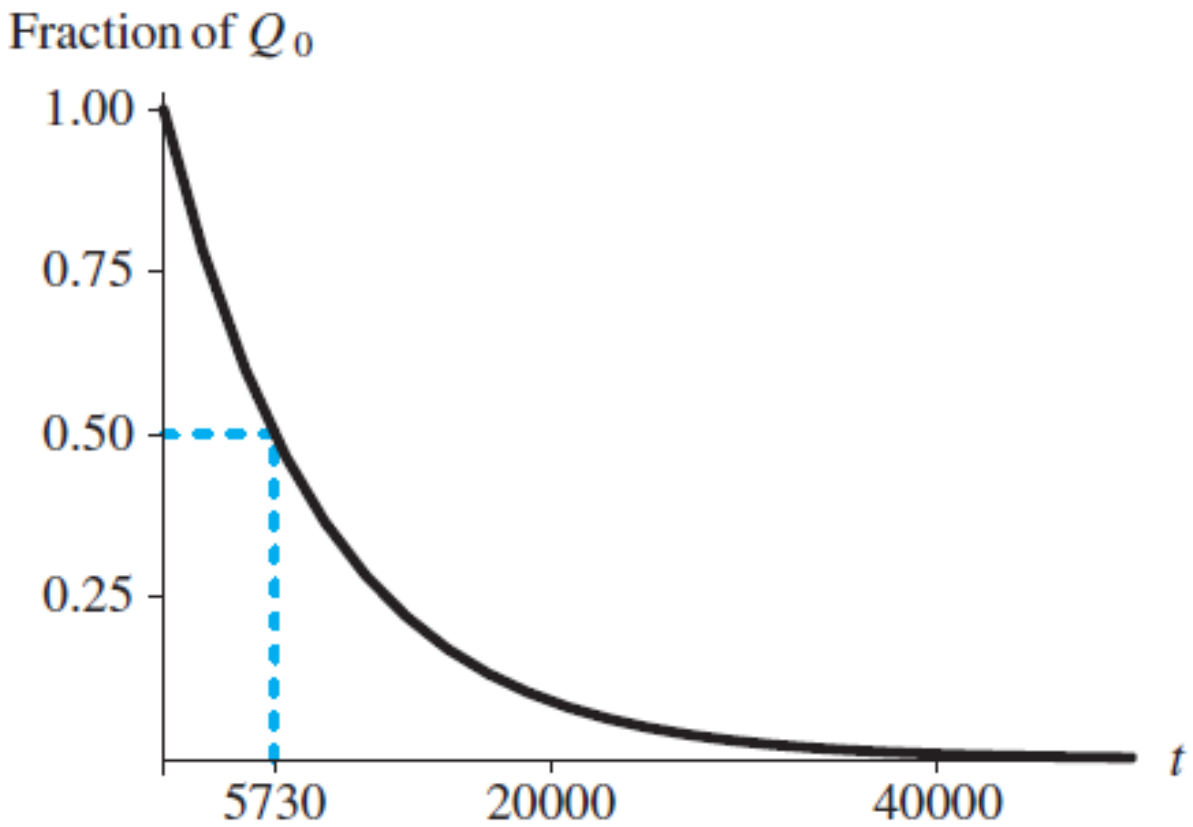
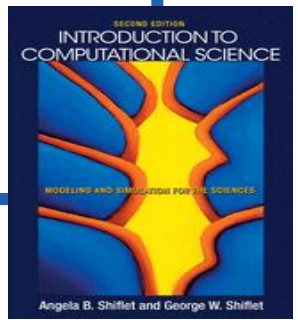


Figure 2.2.5 The half-life of radioactive carbon-14 indicated as 5730 yr

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Quick Review Questions

- Question 5
  - Radium-226 has a continuous decay rate of about 0.0427869% per year. Determine its half-life in whole years.

$$Q = 0.5Q_0$$

$$0.5Q_0 = Q_0 e^{-0.000427869t}$$

$$\ln(0.5) = \ln\left(e^{-0.000427869t}\right) = -0.000427869t$$

$$t = \frac{\ln(0.5)}{-0.000427869} \approx 1619.99 \text{ tahun}$$

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